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Data Visualization of Sephora Beauty Products:
An Evidence-Based Approach to Visual Analytics



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Professor: Neeraj Jha

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Data Visualization of Sephora Beauty Products:
An Evidence-Based Approach to Visual Analytics

Course: Data Visualization
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Declaration:

I hereby declare that this assignment is my own work and has not been submitted, either in whole or in part, for any other academic purpose. All sources of information have been appropriately acknowledged.

Signature: Marcelo da Fonseca Oliveira
Date: 09/04/2026

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1. Introduction

I wanted to look at real retail data for this project. The Sephora dataset from Kaggle has details on 8,494 products and it seemed like a solid way to practice visualization without using made-up numbers. My main goal was simple. Could I answer basic business questions about price and popularity just by looking at charts built from this data? I stuck to basic rules about how people actually see and read graphs. The idea was to keep it clean and let the numbers do the talking.

2. Methodology and Data Structure

The dataset has 27 columns and 8,494 rows. That gave me plenty to work with. I cleaned it up first. Empty review fields became zeros. Empty ratings became zeros. Kelleher and Wagener 2011 talk about data integrity and this just made sense so I could tell which products had zero engagement versus no data at all. I set the file up in Excel with a star schema. The main table holds the product details. Separate tabs hold the brand and category lookups. I used VLOOKUP to connect things. I also added eight new columns to help slice the data differently. One of those columns calculates a Value Score. That is just the rating divided by the price and then adjusted by ten to keep the numbers tidy. It helped me find the items that deliver more stars per dollar spent.

3. Analytical Framework and Techniques

I had ten questions going into this. Does higher price mean better rating? Do limited editions actually perform better? To answer them I leaned on basic Excel. Lots of IF statements to group products into price buckets. Nested formulas to rank popularity. The Value Score metric ended up being the most interesting. You look at the list sorted by Value Score and suddenly cheap drugstore items are sitting near the top next to premium brands. It raises some questions about how we think about price and quality in beauty products.

4. Visualization Design and Theoretical Justification

Most of the charts in this report are bar charts. There is a reason for that. Cleveland and McGill 1984 figured out ages ago that people are best at comparing lengths. Bars lined up on the same baseline are almost impossible to misread. When I had long brand names I just turned the bars sideways so the labels fit without overlapping. I used one pie chart. I know pie charts get a bad rap. Cleveland and McGill rank angle perception pretty low. But Few 2012 points out that for showing parts of a whole they still work fine if the number of slices is small. My goal there was just to show the split between categories not to ask someone to measure a five degree difference. I kept the design quiet. Tufte 1997 talks about maximizing the data to ink ratio. So no weird

gradients or 3D effects. I also avoided red and green together because that is a problem for people with color vision issues.

5. Results and Discussion

So what did I find. First off brand concentration is huge. The top twenty names get most of the attention. The review counts drop off hard after the first few brands. Sephora stocks a ton of products but people tend to stick with what they know. Price varies a lot by category. Fragrance is just expensive across the board. Skincare and makeup are all over the place. You can find a cheap lipstick next to a pricey one on the same page and the data does not tell you which one is actually better. That brings me to the most interesting part. Price and rating are barely related. I expected to see at least some upward trend. There is not much there. A fifteen dollar moisturizer can have a higher average rating than a two hundred dollar cream. The data suggests that customer satisfaction in this category is not something you can just buy. It is about whether the product works for that specific person. Limited edition items get a lot of likes. That makes sense. Scarcity creates buzz. But their actual star ratings are nothing special. They sit right around the average. It seems like people click the heart button because something is rare or has cool packaging. That does not mean they think the formula inside is a miracle. My Value Score calculation confirmed a lot of this. If you are just looking for the best rating per dollar spent budget products absolutely dominate the top of the list. There is real value in the lower price tiers. One last thing I noticed. The super viral products with thousands of likes tend to have a lower ratio of written reviews. It is like a crowd where everyone takes a photo but nobody stops to talk about the experience. The engagement becomes broad but shallow.

6. Conclusion

This project was a good reminder that data often contradicts assumptions. The beauty industry markets luxury and exclusivity hard but the numbers tell a different story. You do not have to spend a lot to get a well rated product. Sticking to simple charts based on Cleveland and McGill's hierarchy made it easy to see those patterns without getting lost in flashy visuals. If I had more time I would want to look at how these ratings shift over time. But for a snapshot of nearly eighty five hundred items the picture is pretty clear. Trust the ratings more than the price tag.

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